A First Course in Abstract Algebra

Section 53: Galois Theory

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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53 Galois Theory

Exercise 53.1: The embedding index of a multiquadratic field

The characteristic is zero, so the extension is separable. Thus its index, the number of embeddings fixing $\mathbb{Q}$, equals its given degree: $\{K : \mathbb{Q}\} = 8$.

Exercise 53.2: The group order of a multiquadratic field

All independent sign changes of $\sqrt{2}, \sqrt{3}, \sqrt{5}$ are automorphisms. They give $2^3 = 8$ maps, exhausting the degree bound. Thus $|G(K/\mathbb{Q})| = 8$.

Exercise 53.3: The subgroup fixing the bottom field

By definition $\lambda(\mathbb{Q}) = G(K/\mathbb{Q})$, so its order is 8. The whole rational Galois group fixes the bottom field.

Exercise 53.4: Fixing two specified radicals

The field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4 over $\mathbb{Q}$. The subgroup fixing it has order $[K : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 8/4 = 2$, consisting of the identity and the sign change of $\sqrt{5}$.

Exercise 53.5: Fixing a product radical

The field $\mathbb{Q}(\sqrt{6})$ has degree 2 over $\mathbb{Q}$, so its fixing subgroup has order $8/2 = 4$. Equivalently the signs of $\sqrt{2}$ and $\sqrt{3}$ must agree, while the sign of $\sqrt{5}$ is arbitrary.

Exercise 53.6: Fixing a triple-product radical

The quadratic field $\mathbb{Q}(\sqrt{30})$ has degree 2 over $\mathbb{Q}$, so its fixing subgroup has order 4. Its sign changes are exactly the choices with a positive product of the three radical signs.

Exercise 53.7: Fixing a sum of two radicals

Put $\alpha = \sqrt{2} + \sqrt{6}$. Its square is $8 + 4\sqrt{3}$, so $\sqrt{3} = (\alpha^2 - 8)/4 \in \mathbb{Q}(\alpha)$. Also $\sqrt{2} = \alpha/(1 + \sqrt{3})$ lies in this field. Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4; its fixing subgroup has order $8/4 = 2$.

Exercise 53.8: Fixing the entire top field

An automorphism fixing every element of K is the identity. Therefore $\lambda(K)$ is the trivial subgroup, of order 1.

Exercise 53.9: The group of fourth roots of unity

The splitting field of $x^4 - 1$ is $\mathbb{Q}(i)$. The two automorphisms send i to i and $-i$. Their composition table is that of the cyclic group $\mathbb{Z}_2$; the nonidentity map is complex conjugation.

Exercise 53.10: A relative finite-field Frobenius

Since $729 = 9^3$, the degree of GF(729) over GF(9) is 3. Its Galois group is cyclic of order 3, generated by $a \mapsto a^9$. The third power is $a \mapsto a^{729} = a$; a smaller positive power cannot fix all 729 elements by the polynomial root bound.

Exercise 53.11: The cubic splitting-field correspondence

Let $r = \sqrt[3]{2}$, $\omega = (-1 + i\sqrt{3})/2$, $K = \mathbb{Q}(r, \omega)$, $\rho(r) = r\omega$, $\rho(\omega) = \omega$, and let κ be complex conjugation.

(a) The six maps and their images are

σ	$\sigma(r)$	$\sigma(i\sqrt{3})$
1	r	$i\sqrt{3}$
ρ	$r\omega$	$i\sqrt{3}$
ρ^2	$r\omega^2$	$i\sqrt{3}$
κ	r	$-i\sqrt{3}$
$\rho\kappa$	$r\omega$	$-i\sqrt{3}$
$\rho^2\kappa$	$r\omega^2$	$-i\sqrt{3}$

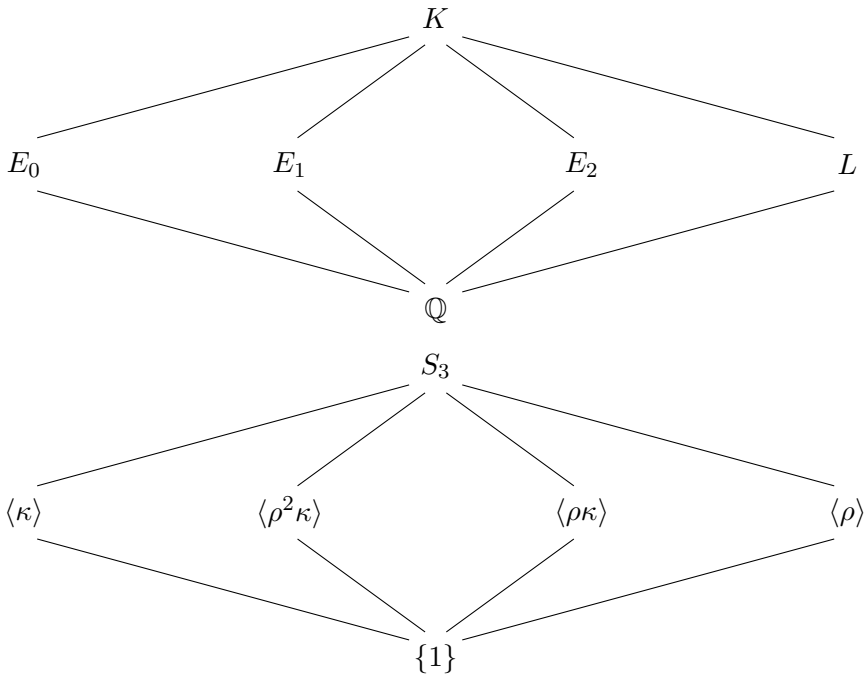
These account for all six rational embeddings of the degree-six splitting field.

(b) Their faithful action on the three cubic roots identifies the group with S_3. In particular $\rho^3 = \kappa^2 = 1$ and $\kappa\rho\kappa = \rho^{-1}$.

(c) The correspondence is

subgroup	fixed field
S_3	$\mathbb{Q}$
$\langle\rho\rangle$	$L = \mathbb{Q}(\omega)$
$\langle\kappa\rangle$	$E_0 = \mathbb{Q}(r)$
$\langle\rho\kappa\rangle$	$E_2 = \mathbb{Q}(r\omega^2)$
$\langle\rho^2\kappa\rangle$	$E_1 = \mathbb{Q}(r\omega)$
$\{1\}$	K

For instance $\rho\kappa(r\omega^2) = r\omega \cdot \omega^{-2} = r\omega^2$. The field degrees (3 for the E_j, 2 for L) and the Galois correspondence prove these are the full fixed fields. The diagrams below have only the displayed covering relations; invert one to obtain the other.



Exercise 53.12: The group of two quadratic factors

Since $x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$, its splitting field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ of degree 4. The independent sign flips of the two radicals commute and each has order 2. Thus its group is $\mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group.

Exercise 53.13: The group of third roots of unity

The splitting field is $\mathbb{Q}(\omega)$ with $\omega^2 + \omega + 1 = 0$. Its two maps send ω to ω and ω^2, so the group is cyclic of order 2.

Exercise 53.14: An abstract group does not determine its field

Take $K_1 = \mathbb{Q}(\sqrt{2})$ and $K_2 = \mathbb{Q}(\sqrt{3})$ over $F = \mathbb{Q}$. Both have Galois group $\mathbb{Z}_2$. They are not isomorphic as fields: any isomorphism fixes $\mathbb{Q}$ and would send $\sqrt{2}$ to a square root of 2 in K_2. Writing $(a + b\sqrt{3})^2 = 2$ forces $ab = 0$, leading to $a^2 = 2$ or $b^2 = 2/3$, neither with rational solutions. Thus such an image does not exist.

Exercise 53.15: Ten statements about the Galois correspondence

Here "finite normal" has Fraleigh's definition: a *separable splitting extension*, often called a finite Galois extension.

(a) False. The subgroup-to-fixed-field correspondence is one-to-one.

(b) False. Inclusion reverses: $E \subseteq L$ implies $\lambda(L) \subseteq \lambda(E)$.

(c) True. The same splitting polynomials work over E, and separability descends along the tower.

(d) True. Each degree equals the order of its Galois group.

(e) True. A normal subgroup corresponds exactly to a normal intermediate extension of F.

(f) False. $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ is simple but has cyclic group of order 4, which has a proper subgroup of order 2.

(g) False. A quadratic separable extension has simple Galois group $\mathbb{Z}_2$.

(h) True. Finite-field Galois groups are cyclic.

(i) False in characteristic 2 with this definition. $\mathbb{Z}_2(t^{1/2})/\mathbb{Z}_2(t)$ has degree 2 but is inseparable, hence is not "finite normal" as Fraleigh defines it (it is normal in the modern splitting-only sense).

(j) True. A quadratic in characteristic different from 2 is separable and splits in its root field.

Exercise 53.16: Abelian Galois extensions in a tower

The group $G(K/E)$ is a subgroup of the abelian group $G(K/F)$, hence is abelian. If E/F is normal, restriction gives $G(E/F) \cong G(K/F)/G(K/E)$. A quotient of an abelian group is abelian. Thus K/E and E/F are both abelian extensions in the requested sense. In fact every subgroup of an abelian group is normal, so the normality of intermediate E/F follows automatically here.

Exercise 53.17: Norm and trace lie in the base field

For any $\tau \in G(K/F)$, left multiplication $\sigma \mapsto \tau\sigma$ permutes the group. Hence

$$\tau\left(\prod_{\sigma}\sigma(\alpha)\right) = \prod_{\sigma}(\tau\sigma)(\alpha) = \prod_{\sigma}\sigma(\alpha),$$

and the analogous equality holds for the sum. Thus norm and trace are fixed by the entire group. Its fixed field is exactly F by the Galois correspondence, so both belong to F.

Exercise 53.18: Eight norm and trace calculations

The four embeddings independently change the signs of $\sqrt{2}, \sqrt{3}$.

(a) $N(\sqrt{2}) = (\sqrt{2})^2(-\sqrt{2})^2 = 4$.

(b) $N(\sqrt{2}+\sqrt{3}) = (\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})(-\sqrt{2}+\sqrt{3})(-\sqrt{2}-\sqrt{3}) = 1$, since the first pair is -1 and the second pair is -1.

(c) $N(\sqrt{6}) = (\sqrt{6})^2(-\sqrt{6})^2 = 36$.

(d) Every embedding fixes 2, so $N(2) = 2^4 = 16$.

(e) The two copies each of $\sqrt{2}, -\sqrt{2}$ sum to 0.

(f) The four sign choices of $\sqrt{2}+\sqrt{3}$ sum to 0.

(g) The two copies each of $\sqrt{6}, -\sqrt{6}$ sum to 0.

(h) The four copies of 2 sum to 8.

Exercise 53.19: Norm and trace from minimal-polynomial coefficients

Because $K = F(\alpha)$ is finite normal, its n automorphisms give exactly the n distinct roots of the minimal polynomial of α. Thus

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = \prod_{\sigma \in G(K/F)} (x - \sigma(\alpha)).$$

Comparing constant terms gives $a_0 = (-1)^n \prod_\sigma \sigma(\alpha)$, so (a) $N_{K/F}(\alpha) = (-1)^n a_0$. Comparing the coefficient of x^{n-1} gives (b) $T_{K/F}(\alpha) = -a_{n-1}$.

Exercise 53.20: The group order divides the factorial

Let Ω be the set of distinct roots of f in its splitting field K, and let $m = |\Omega| \leq n$. Because all irreducible factors are separable, K/F is finite normal in the book's sense. The faithful action of $G(K/F)$ on its generating roots embeds it into S_m. Lagrange's theorem gives $|G(K/F)| \mid m!$, and $m! \mid n!$. Hence the order divides $n!$. Repeated irreducible factors in f cause no difficulty when one uses distinct roots.

Exercise 53.21: The natural permutation group of a polynomial

For $\sigma \in G(K/F)$, its values on the roots define a permutation, since applying σ to $f(\alpha) = 0$ gives $f(\sigma\alpha) = 0$. Restriction respects composition. Its kernel fixes every root, hence fixes the field they generate over F, namely K, so its kernel is trivial. Thus the Galois group is naturally isomorphic to its image, a subgroup of the permutations of the distinct roots.

Exercise 53.22: Cyclotomic extensions over a general field

(a) The powers of a primitive nth root ζ give all roots of $x^n - 1$. Its derivative nx^{n-1} has no common root with it under the characteristic hypothesis. Thus $F(\zeta)$ is a finite separable splitting extension, hence finite normal.

(b) Any automorphism sends ζ to another primitive root ζ^a, where a is invertible modulo n. The map assigning a to that automorphism is injective, and composition multiplies exponents modulo n. It embeds the group into the abelian unit group $(\mathbb{Z}/n\mathbb{Z})^\times$. Therefore the Galois group is abelian, even when not every invertible exponent occurs.

Exercise 53.23: Cyclic extensions and their intermediate fields

(a) A subgroup of a cyclic group is cyclic, so $G(K/E)$ is cyclic. If E/F is normal, its group is the quotient $G(K/F)/G(K/E)$, also cyclic. In fact the normality follows automatically because every subgroup of a cyclic group is normal.

(b) If $|G(K/F)| = n$, a cyclic group has exactly one subgroup of each order dividing n. An intermediate field of degree d over F corresponds to a subgroup of index d, or order n/d. Hence for each divisor d of n there is exactly one such intermediate field.

Exercise 53.24: The polynomial formed from every automorphism

(a) Each $\tau \in G(K/F)$ permutes the factors of $f(x) = \prod_\sigma (x - \sigma\alpha)$, since $\sigma \mapsto \tau\sigma$ permutes the group. Thus every coefficient is fixed by the group, and belongs to F.

(b) All distinct conjugates of α occur, by the extension theorem, and each occurs the same number of times: the maps taking α to a specified conjugate form a coset of its stabilizer $G(K/F(\alpha))$. Thus

$$f(x) = \operatorname{irr}(\alpha, F)^{[K:F(\alpha)]}.$$

Since the minimal polynomial is nonconstant, this equals the minimal polynomial exactly when the exponent is 1, namely when $K = F(\alpha)$.

Exercise 53.25: A compositum corresponds to an intersection

An automorphism fixes the join $E \vee L$ if and only if it fixes both E and L: one direction follows from containment, and the other because it then fixes every rational expression in elements of the two fields. Therefore

$$G(K/(E \vee L)) = G(K/E) \cap G(K/L).$$

This is the reversal of taking the smallest field containing two given fields.

Exercise 53.26: An intersection corresponds to a generated subgroup

Put $H = \langle G(K/E), G(K/L) \rangle$. An element is fixed by H exactly when it is fixed by both generating subgroups, so

$$K_H = K_{G(K/E)} \cap K_{G(K/L)} = E \cap L.$$

The bijective Galois correspondence now gives

$$G(K/(E \cap L)) = \langle G(K/E), G(K/L) \rangle.$$

One must generate a subgroup, rather than merely take the union, which need not be a group.